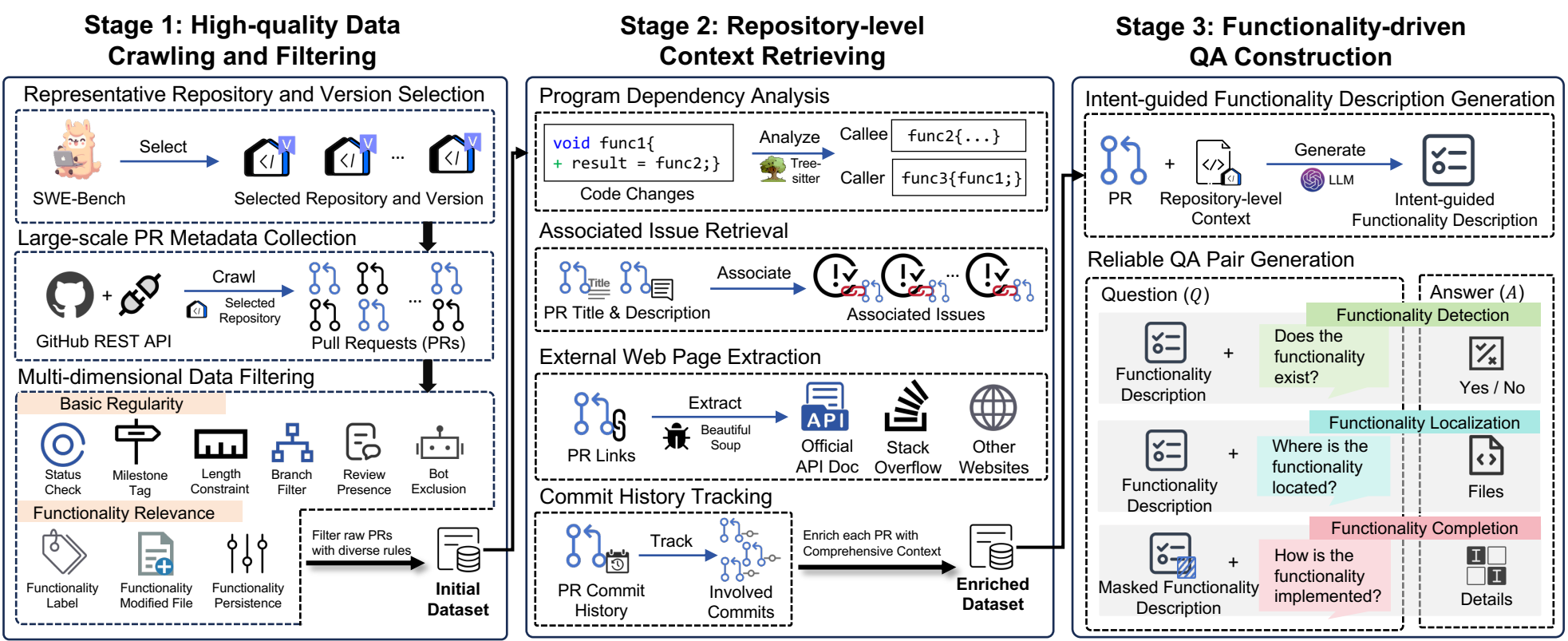
Stage 1: High-quality Data Crawling and Filtering
Representative Repository and Version Selection
SWE-Bench
Select
Selected Repository and Version
Large-scale PR Metadata Collection
GitHub REST API
Crawl
Selected Repository
Pull Requests (PRs)
Multi-dimensional Data Filtering
Basic Regularity
Status Check
Milestone Tag
Length Constraint
Branch Filter
Review Presence
Bot Exclusion
Functionality Relevance
Functionality Label
Functionality Modified File
Functionality Persistence
Filter raw PRs with diverse rules
Initial Dataset
Stage 2: Repository-level Context Retrieving
Program Dependency Analysis
void func1{
+ result = func2;}
Code Changes
Analyze
Tree-sitter
Callee
func2{...}
Caller
func3{func1;}
Associated Issue Retrieval
PR Title & Description
Associate
Associated Issues
External Web Page Extraction
PR Links
Extract
Beautiful Soup
Official API Doc
Stack Overflow
Other Websites
Commit History Tracking
PR Commit History
Track
Involved Commits
Enrich each PR with Comprehensive Context
Enriched Dataset
Stage 3: Functionality-driven QA Construction
Intent-guided Functionality Description Generation
PR
Repository-level Context
Generate
LLM
Intent-guided Functionality Description
Reliable QA Pair Generation
Question (Q)
Answer (A)
Functionality Detection
Functionality Description
Does the functionality exist?
Yes / No
Functionality Localization
Functionality Description
Where is the functionality located?
Files
Functionality Completion
Masked Functionality Description
How is the functionality implemented?
Details